Figure B2: Agency Decision Protocol

DIFFERENTIAL RESPONSE ALLEGED MALTREATMENT GUIDE

Track assignment determined by the presence of imminent danger, level or risk, number of previous reports, source of the report and/or presenting referral characteristics such as type of alleged maltreatment and age of the alleged victim.

Neglect

High Risk Assessment	Family Assessment Response
·Failure to provide medical care in life endangering situations	·Vulnerable child without supervision
·Child inadequately supervised, imminent danger	·Medical diagnosis of failure to thrive
·Child abandonment	·Housing conditions pose health, safety or harm to the children
·Child exposed to methamphetamine manufacturing	·Domestic issues between adults in residence endanger the safety and welfare of the child
·Child access to drugs	·Educational neglect
·Alleged PRAN assaults non-offending victim while holding child	·Child born exposed to chemicals/drugs
·Weapons are being used in the vicinity of the child	·Caregiver involved in possession, use, sale of a controlled substance in the presence of the child
·Child intervenes in an incident of domestic violence and is injured	·Caregiver driving under the influence with child
·Alleged domestic violence PRAN has made suicidal or homicidal statements	·Child has a physical, mental or emotional condition requiring care that is not being received
	·Failure to protect from conditions actions which endanger child
	·Child present or involved with parent committing a criminal act
	·Items thrown in the vicinity of child
	·Child intervenes in an incident of domestic violence and is not injured

Physical

High Risk Assessment	Family Assessment Response
· Life threatening injury	· Threatened physical abuse
·Report of non-accidental injury which requires medical attention	· Present visible injury
·Severe injury(Broken bone, injury to the head, torso or genitals)	· First or Second degree burns (non-accidental)
·Third degree burns (non-accidental)	· Report of non-accidental injury, not currently present
· Physical punishment to child less than 3 y/o involving shaking, throwing or hitting head or trunk	· Unexplained injury
· Excessive physical injuries	· Excessive/extreme/severe punishment without injury
· Unreasonable confinement or restraint including tying, locking, caging or chaining child has injuries	
· Child fatality with other children living in the home	

Sexual

High Risk Assessment	Family Assessment Response
· All allegations of intra-familial sexual abuse	

Emotional

High Risk Assessment	Family Assessment Response
· Child is restrained or restricted from leaving during a domestic violence incident	· Verbal acts or omissions which have an observable adverse effect on the child (name calling, language, constant yelling) unreasonable force or cruelty
· Unreasonable confinement or restraint including tying, locking, caging or chaining or unreasonable force or cruelty	· Caregiver shows little or no attachment
	· Child shows behavioral changes that appear to be related to domestic violence and is fearful of physical injury or death to self or others

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Notes: This image shows the CPS agency's protocol for responding to an allegation of child abuse or neglect. "High-Risk Assessment" designates a higher-priority referral. "Family Assessment Response" designates a lower-priority referral.

Table B1: Redacted Examples of Worker Team Discussions and Corresponding Decisions

Discussion Text	Decision
“Discussion: [Caregiver] arrested for DUI after getting in an accident with unrestrained child in the car."	Screened in (Low Priority)
“Discussion: Firearms in the home and possibly loaded and unlocked. **** has threatened to kill ****** and reached for a firearm. **** has hit ****** and tried to strangle him."	Screened in (High Priority)
“[Algorithm Tool Scores]: 8, 2 and 1 Discussion: Team feels like the family utilized resources like the crisis center. The family could benefit from case management and prevention resources through CCR [Community Response]. A/N [abuse/neglect] does not meet criteria to assign with CPS. No marks/ bruises. Dispo[sition]: FFR [File for Reference]. CCR”	Screened out, voluntary services
“discussion: Concerns for being thrown down the stairs. Cultural situation needs to be assessed and kept in mind. No fear of going home."	Screened out, informal visitor
“No current child protection concerns. Past incidents. RP [reporter] heard ****** has public assistance with housing. Worry since she is sleeping at ****** and she is housing. RP called Alt to violence and they said to call CPS RP heard ****** has criminal [history] and did heroin in past and now smokes MJ. Hig.h risk for verbal abuse. Dad drinks amount [unknown] *** and safe [unknown] if locked. ***** has dementia. He is on **d marriage and past wife DV”	Screened out, no services
“No allegation of abuse or neglect. custody issues. home schooled parents separated for last 3 weeks, kids with dad and his daughter and family (*** and *****) in *****, ** ***** was exposed to ****? [...] and worried she will find **** *********? ******* **** things down and said she doesnt live at moms anymore? yelling screaming cursing and throwing things in the relationship in the past dad said mom is bipolar and needs medication. mom admitted to throwing things in the past dad called mom therapist? dad family is support to the kids dad doesnt agree to ******* getting help?"	Screened out, no services

Notes: Discussions have been redacted. Brackets have been added or substituted for clarification. Listed text is a subset of the full discussion text in some instances. Stars indicate words that were replaced by a de-identification script to anonymize the text data before it was shared with the researchers.

Appendix C: Deviations from Pre-analysis Plan

The experiment was pre-registered on the AEA RCT Registry under ID AEARCTR-0006311.

1. **Randomization Implementation Complications.** Two unforeseen implementation challenges with respect to household randomization required us to adapt our design.

   - Between November 2020-March 2021, a small group of randomization statuses was unintentionally overwritten by our implementing partner. To address this, we omit the $N = 238$ children for whom treatment assignment is unknown (see sample restrictions in Table A.1).
   - In March 2021, the implementing partner attempted to correct the error, but in doing so inadvertently changed the randomization procedure and introduced two complications. First, when a recomposed family was re-referred with at least one child previously assigned to the intervention group and at least one child previously assigned to control, all children in such conflicting-status referrals were assigned to the intervention group. A small ($N = 80$) subset of children in the analysis sample therefore switched from control to intervention status during the trial. We check the robustness of our results to this complication in Table A.5.
   - The March 2021 correction introduced a second challenge: children without a listed biological mother were randomly assigned an intervention status at the individual level instead of at the sibling-group level. Given the concurrent change that assigned referrals with conflicting-status children to the intervention group, beginning in March 2021 larger sibling groups were more likely to be assigned to the intervention group. For example, a group of three siblings without a listed mother was assigned to control with a probability of: $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$ instead of the intended $\frac{1}{2}$. Most children (80 percent) were referred with a listed mother and therefore unaffected. For the $N = 843$ children without a listed mother, treatment assignment remained random conditional on the number of listed children. We therefore include controls for each sibling group size greater than one for children listed without a mother and who were first randomized on or after March 2021. We refer to these controls throughout the paper as "randomization controls."

2. **Suggestive Maltreatment Outcome.** Among ICD codes suggestive of maltreatment, we had included a vague reference to the trauma mortality prediction model (TMPM). However, the inconsistent level of ICD-code detail in our received data, combined with difficulty of crosswalks between ICD-9 to ICD-10, made it impossible to apply the TMPM to our setting. We thus report effects on an alternative measure of suggestive maltreatment provided by Schnitzer et al. (2011) in Appendix Table A.6. As would have been the case for the TMPM, this outcome is only available for a subset of child ages.

3. **Aggregating Hospital Outcomes.** Although our pre-analysis plan listed a set of key medical outcomes using hospital data, we had not pre-specified how we would aggregate effects across different outcomes. Acknowledging this, we report various ways one could aggregate results in Table A.5 and show that our results do not depend on how we aggregate our hospital outcome variables together.

4. **Gender Disparities.** We had pre-registered examining disparities by race, ethnicity, and socioeconomic status. However, given that we also find gender health disparities in our setting, and these disparities were of interest to the agency partner, we therefore also examine effects of the algorithm tool on gender health disparities.